

Name: _____

Period: _____

Date: _____

Topic 1.6 Guided Notes: Sensation and Vision (1 of 2)

Unit 1: Biological Bases of Behavior | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. This worksheet covers Parts 1-2 (Thresholds and Vision) — Hearing, Chemical Senses, and Body Senses are on the second worksheet for this topic.

Source page: appsychreviewhub.com/overviews/topic1.6

Part 1: The Big Picture (Thresholds and Sensory Basics)

Word Bank

<i>Weber's Law</i>	<i>Sensory Interaction</i>	<i>Synesthesia</i>	<i>Difference Threshold</i>
<i>Transduction</i>	<i>Sensory Adaptation</i>	<i>Absolute Threshold</i>	<i>Sensation</i>

- _____: the bottom-up process by which our sensory receptors and nervous system receive and represent stimulus energy from our environment.
- _____: the process by which a sense organ converts that incoming physical stimulus energy into neural impulses the brain can interpret.
- _____: the minimum amount of stimulation needed to detect a particular stimulus (like a light, sound, or smell) 50% of the time.
- _____ (Just Noticeable Difference, or JND): the minimum difference between two stimuli required for a person to detect a change between them 50% of the time.
- _____: the principle that for a difference to be perceived between two stimuli, they must differ by a constant minimum percentage (rather than a constant amount).
- _____: the diminished sensitivity to a constant, unchanging stimulus as a result of continuous stimulation — why you stop noticing a scent after being in a room for a while.
- _____: the principle that one sense can influence another — for example, the smell of food can affect how it tastes.
- _____: a rare condition in which stimulation of one sense (such as hearing) involuntarily triggers an experience in a different sense (such as seeing colors).

Apply It: Which Concept Is This?

Using the word bank above, identify which concept each scenario best illustrates.

#	Scenario	Concept
1	A swimmer can barely feel the cold water on his skin within a few minutes of getting in, even though it felt freezing at first.	
2	Adding one extra candle to a birthday cake with 3 candles is obviously noticeable, but adding one candle to a cake with 50 is not.	
3	A chef says that a beautifully plated dish always seems to taste better than the same food served sloppily on a paper plate.	

Part 2: Vision

Word Bank

<i>Blindsight</i>	<i>Lens</i>	<i>Opponent-Process Theory</i>	<i>Prosopagnosia</i>
<i>Cornea</i>	<i>Pupil</i>	<i>Rods</i>	<i>Nearsightedness</i>
<i>Monochromatism</i>	<i>Young-Helmholtz Trichromatic Theory</i>	<i>Iris</i>	<i>Color Vision Deficiency</i>
<i>Farsightedness</i>	<i>Dichromatism</i>	<i>Cones</i>	<i>Retina</i>

The Path of Light Through the Eye

- _____: the clear, protective outer layer at the front of the eye that first bends light to begin the focusing process.
- _____: the small, adjustable opening in the center of the eye that controls how much light enters.
- _____: the colored, ring-shaped muscle that adjusts to control the size of the pupil.
- _____: the transparent structure behind the pupil that changes shape to focus incoming light onto the retina.
- _____: the light-sensitive inner surface of the eye, containing millions of receptor cells, where the image actually gets focused.

When the Eye Doesn't Focus Correctly

- _____ (Myopia): a condition where nearby objects are seen clearly, but distant objects appear blurry, because the image focuses in front of the retina.
- _____ (Hyperopia): a condition where distant objects are seen clearly, but nearby objects appear blurry, because the image focuses behind the retina.

The Retina's Receptors

- _____: retinal receptor cells that are highly sensitive to dim light and are responsible for peripheral and twilight (black-and-white) vision.
- _____: retinal receptor cells, concentrated near the center of the retina, that function best in daylight or well-lit conditions and detect fine detail and color.

Theories of Color Vision

- _____: the theory that the retina contains three different types of color receptors (cones) — red, green, and blue — and that combinations of their activity produce the full spectrum of color we see.
- _____: the theory that opposing retinal processes (red-green, blue-yellow, black-white) enable color vision, and that this opposition explains why staring at one color produces an afterimage of its opposite.

When Vision Breaks Down

- _____: the inability to distinguish between certain colors, most commonly red and green.
- _____: a form of color vision deficiency in which a person can only perceive two of the three primary colors.
- _____: a rare condition in which a person sees no color at all, perceiving the world only in shades of black, white, and gray.

- _____: a neurological disorder, unrelated to color vision, that impairs a person's ability to perceive or recognize faces — also known as face blindness.
- _____: a condition in which a person who is cortically blind (due to brain damage rather than eye damage) can still respond to visual stimuli they report not consciously seeing.

Apply It: Vision Structure or Disorder?

#	Scenario	Structure/Disorder
1	A student can read the whiteboard perfectly but holds books very close to her face to read them clearly.	
2	A patient with damage to a specific visual-processing brain region insists he cannot see anything, yet he can accurately catch a ball thrown at him.	
3	A patient recognizes his wife's voice instantly but stares blankly at her face, unable to recognize her by sight alone.	



Video Notes: Sensation and Perception (Crash Course Psychology #5)

Answer the questions below as you watch — some ideas will already be familiar from Parts 1-2, and that's fine; reinforcing them is the point.

- According to the video, what is the difference between sensation and perception?
- What example does the video give to explain Signal Detection Theory?
- According to the video, what determines whether we see short vs. long wavelengths of light as different colors, and what does the amplitude of a light wave affect?
- According to the video, how does the Trichromatic Theory differ from the Opponent-Process Theory of color vision?
- What are feature detectors, and what example does the video use to explain parallel processing?

3. Don't Trip Up! (Common Misconceptions)

There is a small area on your retina with no photoreceptors at all — your blind spot, located where the optic nerve exits the eye. You don't normally notice it because your brain fills in the missing information using surrounding visual data.

The Trichromatic and Opponent-Process theories are NOT competing, contradictory ideas — modern research shows both are correct, just operating at different points along the visual pathway. Trichromatic processing happens first, at the level of the cones in the retina; opponent-processing happens second, further along the visual pathway.

4. Quick Check: Review Questions

Circle the best answer for each question below.

1. A researcher determines the softest possible whisper a participant can detect exactly half the time in a silent room. She is measuring the participant's:
 - A. Difference Threshold
 - B. Weber's Law
 - C. Absolute Threshold
 - D. Sensory Adaptation
2. A person holding a 5-pound bag can easily notice when 1 pound is added, but a person holding a 100-pound bag cannot notice that same 1-pound addition. This phenomenon, where a constant percentage rather than a constant amount is needed to notice a difference, is called:
 - A. Absolute Threshold
 - B. Sensory Interaction
 - C. Sensory Adaptation
 - D. Weber's Law
3. A photographer walks from a bright, sunlit sidewalk into a dark movie theater and needs several minutes before he can see the seats around him clearly. Which retinal receptor is primarily responsible for this dark adaptation process?
 - A. The Iris
 - B. The Lens
 - C. Rods
 - D. Cones
4. A graphic designer can distinguish extremely fine details and vibrant colors while working under bright studio lighting, thanks to receptors concentrated near the center of her retina. Which receptor is she primarily relying on?
 - A. The Cornea
 - B. The Pupil
 - C. Rods
 - D. Cones
5. After staring at a bright green square for 30 seconds and then looking at a blank white wall, a viewer sees a reddish afterimage in the same shape. Which theory of color vision best explains this specific afterimage effect?
 - A. Volley Theory
 - B. Frequency Theory
 - C. Opponent-Process Theory
 - D. Young-Helmholtz Trichromatic Theory



Check Your Progress

Done with your notes? Head over to the [Quiz](https://appspsychreviewhub.com/quiz) to test what you've learned:

<https://appspsychreviewhub.com/quiz>